

TCS CodeVita & NQT Advanced: 10-Year Problem Archive

Source-linked, de-duplicated, and paraphrased. Built locally from public archives, official sample pages, and shift-wise community indexes.

Generated on 2026-04-20

Important note: this PDF intentionally avoids reproducing long verbatim copyrighted problem statements. Use the embedded source links for the original wording and samples.

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Method and Coverage

This dossier targets advanced TCS CodeVita questions plus publicly verifiable TCS NQT Advanced Coding problems. CodeVita coverage spans 2018 through the current public Season 13 sample page. For NQT, the earliest stable public evidence I found for the advanced coding section was the 2023 cycle; older NQT years are therefore treated as a gap rather than guessed.

Total archived entries: **65**. Entries marked as title-level catalog records preserve verified year/title/source without copying the full prompt.

Track / Year	Count
TCS CodeVita 2018	9
TCS CodeVita 2019	18
TCS CodeVita 2020 / Season 9	6
TCS CodeVita 2023 / Season 11	7
TCS CodeVita 2024 / Season 12	6
TCS CodeVita 2025-2026 / Season 13	7
TCS NQT Advanced Coding 2023	4
TCS NQT Advanced Coding 2024 / 03 May Shift 1	1
TCS NQT Advanced Coding 2024 / 03 May Shift 2	1
TCS NQT Advanced Coding 2024 / 26 Apr Shift 1	2
TCS NQT Advanced Coding 2024 / 29 Apr Shift 1	1
TCS NQT Advanced Coding 2024 / 29 Apr Shift 2	1
TCS NQT Advanced Coding 2025 / 03 Oct Shift 2	1
TCS NQT Advanced Coding 2025 / 04 Oct Shift 2	1

Explicit exclusions: unstable forum-only 2026 NQT problem recollections, and any entry where a year/story match could not be tied back to a stable public source. This keeps the archive defensible even when it is not fully exhaustive.

Problem Archive

01. Bank Compare - TCS CodeVita 2018

Synopsis. Title/year archived from a public, year-labeled problem repository. Full wording remains at the linked source.

Track: TCS CodeVita

Year / Season: 2018

Verification: Year-labeled GitHub archive with per-problem PDFs and/or solution files under the 2018 directory.

Difficulty Filter: Included because CodeVita is an advanced programming contest; this entry is title-level only in the compiled archive.

Constraints: Source-linked only.

Input Format: Source-linked only.

Output Format: Source-linked only.

Sample I/O: Source-linked only.

Archive Note: This build keeps the verified title/year and source path, but not the full problem text.

Primary sources

- <https://github.com/KAgarwalCodeBase/TCS-Codevita/tree/master/2018>

02. Bird Hunt - TCS CodeVita 2018

Synopsis. Title/year archived from a public, year-labeled problem repository. Full wording remains at the linked source.

Track: TCS CodeVita

Year / Season: 2018

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Primary sources

- <https://github.com/KAgarwalCodeBase/TCS-Codevita/tree/master/2018>

03. Bride Hunting - TCS CodeVita 2018

Synopsis. Title/year archived from a public, year-labeled problem repository. Full wording remains at the linked source.

Track: TCS CodeVita

Year / Season: 2018

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Primary sources

- <https://github.com/KAgarwalCodeBase/TCS-Codevita/tree/master/2018>

04. Chakravyuha - TCS CodeVita 2018

Synopsis. Title/year archived from a public, year-labeled problem repository. Full wording remains at the linked source.

Track: TCS CodeVita

Year / Season: 2018

Verification: Year-labeled GitHub archive with per-problem PDFs and/or solution files under the 2018 directory.

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Primary sources

- <https://github.com/KAgarwalCodeBase/TCS-Codevita/tree/master/2018>

05. Colliding Cannon - TCS CodeVita 2018

Synopsis. Title/year archived from a public, year-labeled problem repository. Full wording remains at the linked source.

Track: TCS CodeVita

Year / Season: 2018

Verification: Year-labeled GitHub archive with per-problem PDFs and/or solution files under the 2018 directory.

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Sample I/O: Source-linked only.

Archive Note: This build keeps the verified title/year and source path, but not the full problem text.

Primary sources

- <https://github.com/KAgarwalCodeBase/TCS-Codevita/tree/master/2018>

06. Cross Words - TCS CodeVita 2018

Synopsis. Title/year archived from a public, year-labeled problem repository. Full wording remains at the linked source.

Track: TCS CodeVita

Year / Season: 2018

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Archive Note: This build keeps the verified title/year and source path, but not the full problem text.

Primary sources

- <https://github.com/KAgarwalCodeBase/TCS-Codevita/tree/master/2018>

07. Jurassic Park - TCS CodeVita 2018

Synopsis. Title/year archived from a public, year-labeled problem repository. Full wording remains at the linked source.

Track: TCS CodeVita

Year / Season: 2018

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Difficulty Filter: Included because CodeVita is an advanced programming contest; this entry is title-level only in the compiled archive.

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Output Format: Source-linked only.

Sample I/O: Source-linked only.

Archive Note: This build keeps the verified title/year and source path, but not the full problem text.

Primary sources

- <https://github.com/KAgarwalCodeBase/TCS-Codevita/tree/master/2018>

08. Skate Board - TCS CodeVita 2018

Synopsis. Title/year archived from a public, year-labeled problem repository. Full wording remains at the linked source.

Track: TCS CodeVita

Year / Season: 2018

Verification: Year-labeled GitHub archive with per-problem PDFs and/or solution files under the 2018 directory.

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Archive Note: This build keeps the verified title/year and source path, but not the full problem text.

Primary sources

- <https://github.com/KAgarwalCodeBase/TCS-Codevita/tree/master/2018>

09. String Rotation - TCS CodeVita 2018

Synopsis. Title/year archived from a public, year-labeled problem repository. Full wording remains at the linked source.

Track: TCS CodeVita

Year / Season: 2018

Verification: Year-labeled GitHub archive with per-problem PDFs and/or solution files under the 2018 directory.

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Sample I/O: Source-linked only.

Archive Note: This build keeps the verified title/year and source path, but not the full problem text.

Primary sources

- <https://github.com/KAgarwalCodeBase/TCS-Codevita/tree/master/2018>

10. Death Battle - TCS CodeVita 2019

Synopsis. Title/year archived from a public, year-labeled problem repository. Full wording remains at the linked source.

Track: TCS CodeVita

Year / Season: 2019

Verification: Year-labeled GitHub archive with round-wise PDFs and solution files for 2019 CodeVita.

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Primary sources

- <https://github.com/KAgarwalCodeBase/TCS-Codevita/tree/master/2019%20round%201%20tcs%20codevita>
- <https://github.com/KAgarwalCodeBase/TCS-Codevita/tree/master/Mock%20Vita%202%202019/MockVita%202%202019>

11. Holes And Balls - TCS CodeVita 2019

Synopsis. Title/year archived from a public, year-labeled problem repository. Full wording remains at the linked source.

Track: TCS CodeVita

Year / Season: 2019

Verification: Year-labeled GitHub archive with round-wise PDFs and solution files for 2019 CodeVita.

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- <https://github.com/KAgarwalCodeBase/TCS-Codevita/tree/master/Mock%20Vita%202%202019/MockVita%202%202019>

12. Marathon Winner - TCS CodeVita 2019

Synopsis. Title/year archived from a public, year-labeled problem repository. Full wording remains at the linked source.

Track: TCS CodeVita

Year / Season: 2019

Verification: Year-labeled GitHub archive with round-wise PDFs and solution files for 2019 CodeVita.

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- <https://github.com/KAgarwalCodeBase/TCS-Codevita/tree/master/Mock%20Vita%202%202019/MockVita%202%202019>

13. Bottlenecks - TCS CodeVita 2019

Synopsis. Title/year archived from a public, year-labeled problem repository. Full wording remains at the linked source.

Track: TCS CodeVita

Year / Season: 2019

Verification: Year-labeled GitHub archive with round-wise PDFs and solution files for 2019 CodeVita.

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- <https://github.com/KAgarwalCodeBase/TCS-Codevita/tree/master/Mock%20Vita%202%202019/MockVita%202%202019>

14. Clock Angle - TCS CodeVita 2019

Synopsis. Title/year archived from a public, year-labeled problem repository. Full wording remains at the linked source.

Track: TCS CodeVita

Year / Season: 2019

Verification: Year-labeled GitHub archive with round-wise PDFs and solution files for 2019 CodeVita.

Difficulty Filter: Included because CodeVita is an advanced programming contest; this entry is title-level only in the compiled archive.

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- <https://github.com/KAgarwalCodeBase/TCS-Codevita/tree/master/Mock%20Vita%202%202019/MockVita%202%202019>

15. Crossword - TCS CodeVita 2019

Synopsis. Title/year archived from a public, year-labeled problem repository. Full wording remains at the linked source.

Track: TCS CodeVita

Year / Season: 2019

Verification: Year-labeled GitHub archive with round-wise PDFs and solution files for 2019 CodeVita.

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- <https://github.com/KAgarwalCodeBase/TCS-Codevita/tree/master/Mock%20Vita%202%202019/MockVita%202%202019>

16. Divine Divisor - TCS CodeVita 2019

Synopsis. Title/year archived from a public, year-labeled problem repository. Full wording remains at the linked source.

Track: TCS CodeVita

Year / Season: 2019

Verification: Year-labeled GitHub archive with round-wise PDFs and solution files for 2019 CodeVita.

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17. Friend Circle - TCS CodeVita 2019

Synopsis. Title/year archived from a public, year-labeled problem repository. Full wording remains at the linked source.

Track: TCS CodeVita

Year / Season: 2019

Verification: Year-labeled GitHub archive with round-wise PDFs and solution files for 2019 CodeVita.

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- <https://github.com/KAgarwalCodeBase/TCS-Codevita/tree/master/Mock%20Vita%202%202019/MockVita%202%202019>

18. Island - TCS CodeVita 2019

Synopsis. Title/year archived from a public, year-labeled problem repository. Full wording remains at the linked source.

Track: TCS CodeVita

Year / Season: 2019

Verification: Year-labeled GitHub archive with round-wise PDFs and solution files for 2019 CodeVita.

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- <https://github.com/KAgarwalCodeBase/TCS-Codevita/tree/master/Mock%20Vita%202%202019/MockVita%202%202019>

19. Lazy Student - TCS CodeVita 2019

Synopsis. Title/year archived from a public, year-labeled problem repository. Full wording remains at the linked source.

Track: TCS CodeVita

Year / Season: 2019

Verification: Year-labeled GitHub archive with round-wise PDFs and solution files for 2019 CodeVita.

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- <https://github.com/KAgarwalCodeBase/TCS-Codevita/tree/master/Mock%20Vita%202%202019/MockVita%202%202019>

20. Lexi String - TCS CodeVita 2019

Synopsis. Title/year archived from a public, year-labeled problem repository. Full wording remains at the linked source.

Track: TCS CodeVita

Year / Season: 2019

Verification: Year-labeled GitHub archive with round-wise PDFs and solution files for 2019 CodeVita.

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- <https://github.com/KAgarwalCodeBase/TCS-Codevita/tree/master/Mock%20Vita%202%202019/MockVita%202%202019>

21. ODI Score - TCS CodeVita 2019

Synopsis. Title/year archived from a public, year-labeled problem repository. Full wording remains at the linked source.

Track: TCS CodeVita

Year / Season: 2019

Verification: Year-labeled GitHub archive with round-wise PDFs and solution files for 2019 CodeVita.

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- <https://github.com/KAgarwalCodeBase/TCS-Codevita/tree/master/Mock%20Vita%202%202019/MockVita%202%202019>

22. Pattern Printing - TCS CodeVita 2019

Synopsis. Title/year archived from a public, year-labeled problem repository. Full wording remains at the linked source.

Track: TCS CodeVita

Year / Season: 2019

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- <https://github.com/KAgarwalCodeBase/TCS-Codevita/tree/master/Mock%20Vita%202%202019/MockVita%202%202019>

23. Prime Faces - TCS CodeVita 2019

Synopsis. Title/year archived from a public, year-labeled problem repository. Full wording remains at the linked source.

Track: TCS CodeVita

Year / Season: 2019

Verification: Year-labeled GitHub archive with round-wise PDFs and solution files for 2019 CodeVita.

Difficulty Filter: Included because CodeVita is an advanced programming contest; this entry is title-level only in the compiled archive.

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- <https://github.com/KAgarwalCodeBase/TCS-Codevita/tree/master/Mock%20Vita%202%202019/MockVita%202%202019>

24. Salary Paid - TCS CodeVita 2019

Synopsis. Title/year archived from a public, year-labeled problem repository. Full wording remains at the linked source.

Track: TCS CodeVita

Year / Season: 2019

Verification: Year-labeled GitHub archive with round-wise PDFs and solution files for 2019 CodeVita.

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- <https://github.com/KAgarwalCodeBase/TCS-Codevita/tree/master/Mock%20Vita%202%202019/MockVita%202%202019>

25. Similar Char - TCS CodeVita 2019

Synopsis. Title/year archived from a public, year-labeled problem repository. Full wording remains at the linked source.

Track: TCS CodeVita

Year / Season: 2019

Verification: Year-labeled GitHub archive with round-wise PDFs and solution files for 2019 CodeVita.

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Archive Note: Some 2019 PDFs were machine-readable only in part, so the compiled archive records title/year/source without restating the full story text.

Primary sources

- <https://github.com/KAgarwalCodeBase/TCS-Codevita/tree/master/2019%20round%201%20tcs%20codevita>
- <https://github.com/KAgarwalCodeBase/TCS-Codevita/tree/master/Mock%20Vita%202%202019/MockVita%202%202019>

26. Uncertain Step - TCS CodeVita 2019

Synopsis. Title/year archived from a public, year-labeled problem repository. Full wording remains at the linked source.

Track: TCS CodeVita

Year / Season: 2019

Verification: Year-labeled GitHub archive with round-wise PDFs and solution files for 2019 CodeVita.

Difficulty Filter: Included because CodeVita is an advanced programming contest; this entry is title-level only in the compiled archive.

Constraints: Source-linked only.

Input Format: Source-linked only.

Output Format: Source-linked only.

Sample I/O: Source-linked only.

Archive Note: Some 2019 PDFs were machine-readable only in part, so the compiled archive records title/year/source without restating the full story text.

Primary sources

- <https://github.com/KAgarwalCodeBase/TCS-Codevita/tree/master/2019%20round%201%20tcs%20codevita>
- <https://github.com/KAgarwalCodeBase/TCS-Codevita/tree/master/Mock%20Vita%202%202019/MockVita%202%202019>

27. Work Life - TCS CodeVita 2019

Synopsis. Title/year archived from a public, year-labeled problem repository. Full wording remains at the linked source.

Track: TCS CodeVita

Year / Season: 2019

Verification: Year-labeled GitHub archive with round-wise PDFs and solution files for 2019 CodeVita.

Difficulty Filter: Included because CodeVita is an advanced programming contest; this entry is title-level only in the compiled archive.

Constraints: Source-linked only.

Input Format: Source-linked only.

Output Format: Source-linked only.

Sample I/O: Source-linked only.

Archive Note: Some 2019 PDFs were machine-readable only in part, so the compiled archive records title/year/source without restating the full story text.

Primary sources

- <https://github.com/KAgarwalCodeBase/TCS-Codevita/tree/master/2019%20round%201%20tcs%20codevita>
- <https://github.com/KAgarwalCodeBase/TCS-Codevita/tree/master/Mock%20Vita%202%202019/MockVita%202%202019>

28. Collecting Candies - TCS CodeVita 2020 / Season 9

Synopsis. Merge candy boxes with minimum total time; each merge costs the sum of candies in the two chosen boxes.

Track: TCS CodeVita

Year / Season: 2020 / Season 9

Verification: PrepInsta explicitly marks this as a TCS CodeVita sample problem from the 2020 Season 9 cycle.

Difficulty Filter: Advanced greedy / heap-style optimization problem.

Constraints: Multiple test cases; each test case provides box counts and candy totals.

Input Format: T test cases, then N and the N box candy counts for each case.

Output Format: Minimum total time required to consolidate all candies into one box.

Sample I/O: See linked Season 9 sample page.

Primary sources

- <https://prepinsta.com/tcs-codevita/python-program-for-collecting-candies-problem/>

29. Counting Rock Samples - TCS CodeVita 2020 / Season 9

Synopsis. Classify a very large stream of rock samples into lab-accepted ppm ranges and return the counts per interval.

Track: TCS CodeVita

Year / Season: 2020 / Season 9

Verification: PrepInsta explicitly marks this as a TCS CodeVita 2020 Season 9 sample problem.

Difficulty Filter: Implementation-heavy counting and range aggregation problem.

Constraints: Large sample counts; intended for linear counting rather than nested scans.

Input Format: Range definitions and rock-sample values as described in the linked sample page.

Output Format: Counts of samples per laboratory range.

Sample I/O: See linked Season 9 sample page.

Primary sources

● <https://prepinsta.com/tcs-codevita/cpp-program-for-counting-rock-sample/>

30. Houses Problem - TCS CodeVita 2020 / Season 9

Synopsis. Maximize the total stolen value from houses in a line without robbing adjacent houses.

Track: TCS CodeVita

Year / Season: 2020 / Season 9

Verification: PrepInsta explicitly marks this as a TCS CodeVita sample problem from Season 9.

Difficulty Filter: Classic dynamic-programming selection problem.

Constraints: Linear DP with house values.

Input Format: House count followed by house values.

Output Format: Maximum obtainable loot under the adjacency restriction.

Sample I/O: See linked Season 9 sample page.

Primary sources

- <https://prepinsta.com/tcs-codevita/java-program-for-houses-problem/>

31. K-th Largest Factor of N - TCS CodeVita 2020 / Season 9

Synopsis. Find the k-th largest factor of a positive integer, or return 1 if the factor count is too small.

Track: TCS CodeVita

Year / Season: 2020 / Season 9

Verification: PrepInsta explicitly labels this as a TCS CodeVita Season 9 sample problem.

Difficulty Filter: Number theory with divisor enumeration under tight constraints.

Constraints: N can be large enough to require efficient factor search.

Input Format: Positive integer N and factor index k.

Output Format: The k-th largest factor, or 1 when it does not exist.

Sample I/O: See linked Season 9 sample page.

Primary sources

- <https://prepinsta.com/tcs-codevita/java-program-for-kth-largest-factor-of-n/>

32. Maneuvering a Cave - TCS CodeVita 2020 / Season 9

Synopsis. Count the number of valid paths from the top-left to the bottom-right of a grid while moving only right or down.

Track: TCS CodeVita

Year / Season: 2020 / Season 9

Verification: PrepInsta explicitly labels this as a TCS CodeVita 2020 Season 9 sample problem.

Difficulty Filter: Combinatorics / DP grid-path counting problem.

Constraints: Multiple test cases over grid dimensions.

Input Format: T test cases, each with row and column counts.

Output Format: Number of valid paths for each test case.

Sample I/O: See linked Season 9 sample page.

Primary sources

- <https://prepinsta.com/tcs-codevita/python-program-for-maneuvering-a-cave/>

33. Staircase Problem - TCS CodeVita 2020 / Season 9

Synopsis. Count the number of ways to climb n stairs when each move can take 1 or 2 steps.

Track: TCS CodeVita

Year / Season: 2020 / Season 9

Verification: PrepInsta explicitly labels this as a TCS CodeVita 2020 Season 9 sample problem.

Difficulty Filter: Classic recurrence / DP problem.

Constraints: Single integer n.

Input Format: One integer representing the staircase height.

Output Format: Number of distinct ways to reach the top.

Sample I/O: See linked Season 9 sample page.

Primary sources

- <https://prepinsta.com/tcs-codevita/python-program-for-staircase-problem/>

34. Antikythera Mechanism - TCS CodeVita 2023 / Season 11

Synopsis. Analyze a coupled gear system from positional and size data to determine the rotation effect on the last gear.

Track: TCS CodeVita

Year / Season: 2023 / Season 11

Verification: PrepInsta explicitly labels it as a sample problem for TCS CodeVita Season 11.

Difficulty Filter: Geometry / graph connectivity / ratio propagation.

Constraints: Source-linked only.

Input Format: Gear positions and radii as described on the linked page.

Output Format: Rotation count or reachability result for the terminal gear.

Sample I/O: See linked Season 11 sample page.

Primary sources

● <https://prepinsta.com/tcs-codevita/java-code-for-antikythera-mechanism-tcs-codevita-prepinsta/>

35. HelpMLA - TCS CodeVita 2023 / Season 11

Synopsis. Publicly indexed Season 11 sample title; consult the linked page for the original story and full I/O contract.

Track: TCS CodeVita

Year / Season: 2023 / Season 11

Verification: PrepInsta explicitly labels it as a sample problem for TCS CodeVita Season 11.

Difficulty Filter: Included as a season-verified advanced CodeVita sample title.

Constraints: Source-linked only.

Input Format: Source-linked only.

Output Format: Source-linked only.

Sample I/O: Source-linked only.

Archive Note: Title is season-verified; this compiled archive does not restate the full prompt.

Primary sources

- <https://prepinsta.com/tcs-codevita/java-code-for-helpmla-tcs-codevita-prepinsta/>

36. No More Symbols - TCS CodeVita 2023 / Season 11

Synopsis. Interpret arithmetic expressions encoded fully in lowercase words, with digits spoken separately and joined by a marker character.

Track: TCS CodeVita

Year / Season: 2023 / Season 11

Verification: PrepInsta explicitly labels it as a sample problem for TCS CodeVita Season 11.

Difficulty Filter: String parsing plus expression evaluation.

Constraints: Source-linked only.

Input Format: Word-based encoded expression in the format described on the linked page.

Output Format: Computed numeric result or the required validation failure output.

Sample I/O: See linked Season 11 sample page.

Primary sources

- <https://prepinsta.com/tcs-codevita/java-code-for-no-more-symbols-tcs-codevita-prepinsta/>

37. Online Shopping - TCS CodeVita 2023 / Season 11

Synopsis. Use missed token claims and a one-time consecutive-day recovery offer to maximize a festive-sale token total.

Track: TCS CodeVita

Year / Season: 2023 / Season 11

Verification: PrepInsta explicitly labels it as a sample problem for TCS CodeVita Season 11.

Difficulty Filter: Sliding-window / range optimization problem.

Constraints: Token sequence and recovery window length.

Input Format: Daily token-claim history plus the special recovery-window parameter.

Output Format: Maximum collectible token value under the offer rules.

Sample I/O: See linked Season 11 sample page.

Primary sources

- <https://prepinsta.com/tcs-codevita/java-code-for-online-shopping-tcs-codevita-prepinsta/>

38. PrimeConstruction - TCS CodeVita 2023 / Season 11

Synopsis. Given distinct integers, find the smallest prime p such that division by every value except the smallest leaves the smallest value as remainder.

Track: TCS CodeVita

Year / Season: 2023 / Season 11

Verification: PrepInsta explicitly labels it as a sample problem for TCS CodeVita Season 11.

Difficulty Filter: Constructive number theory problem.

Constraints: $n < 11$ and $p < 10^{10}$ on the linked sample page.

Input Format: Distinct natural numbers where the minimum acts as q .

Output Format: Smallest valid prime p or None.

Sample I/O: Example on the linked page uses 3 4 5 1 and outputs 61.

Primary sources

● <https://prepinsta.com/tcs-codevita/c-code-for-primeconstruction-tcs-codevita-prepinsta/>

39. The Relationship - TCS CodeVita 2023 / Season 11

Synopsis. Season-verified sample title; the linked page should be consulted for the original statement and full I/O contract.

Track: TCS CodeVita

Year / Season: 2023 / Season 11

Verification: PrepInsta explicitly labels it as a sample problem for TCS CodeVita Season 11.

Difficulty Filter: Included as a season-verified advanced CodeVita sample title.

Constraints: Source-linked only.

Input Format: Source-linked only.

Output Format: Source-linked only.

Sample I/O: Source-linked only.

Archive Note: Title is season-verified; full prompt intentionally stays at source.

Primary sources

- <https://prepinsta.com/tcs-codevita/c-code-for-the-relationship-tcs-codevita-prepinsta/>

40. Worker - TCS CodeVita 2023 / Season 11

Synopsis. Resource-to-task allocation problem indexed as a Season 11 sample.

Track: TCS CodeVita

Year / Season: 2023 / Season 11

Verification: PrepInsta explicitly labels it as a sample problem for TCS CodeVita Season 11.

Difficulty Filter: Greedy / scheduling-style advanced contest problem.

Constraints: Source-linked only.

Input Format: Source-linked only.

Output Format: Source-linked only.

Sample I/O: Source-linked only.

Archive Note: The page verifies the sample-season link; consult it for original wording.

Primary sources

- <https://prepinsta.com/tcs-codevita/cpp-code-for-worker-tcs-codevita-prepinsta/>

41. Best Bubble - TCS CodeVita 2024 / Season 12

Synopsis. Find the minimum swaps required to make an array beautiful by sorting it either ascending or descending.

Track: TCS CodeVita

Year / Season: 2024 / Season 12

Verification: Season 12 GitHub repository README lists it as a MockVita problem.

Difficulty Filter: Permutation / swap-count optimization.

Constraints: N and the array values are provided on the linked repository README.

Input Format: Array length followed by the array.

Output Format: Minimum swap count to reach either ascending or descending order.

Sample I/O: See repository README.

Primary sources

● https://github.com/tisha555/TCS_CodeVita_Season12

42. Good String - TCS CodeVita 2024 / Season 12

Synopsis. Convert a student's name into a nearest-character good name using a provided good string and sum the total ASCII distance.

Track: TCS CodeVita

Year / Season: 2024 / Season 12

Verification: Season 12 GitHub repository README lists it as a MockVita problem.

Difficulty Filter: String transformation with tie-breaking logic.

Constraints: Two input strings: the good string and the student name.

Input Format: Good string on the first line, student name on the second line.

Output Format: Total conversion distance.

Sample I/O: See linked repository / season sample page.

Primary sources

- https://github.com/tisha555/TCS_CodeVita_Season12
- <https://prepinsta.com/tcs-codevita/python-code-for-good-string-tcs-codevita-prepinsta/>

43. Orchard - TCS CodeVita 2024 / Season 12

Synopsis. Compare two fruit rows by counting how many ways each side can choose three non-adjacent trees.

Track: TCS CodeVita

Year / Season: 2024 / Season 12

Verification: Season 12 GitHub repository README lists it as a MockVita problem.

Difficulty Filter: Combinatorial counting / validation problem.

Constraints: Two strings over L and M.

Input Format: Ashok's row then Anand's row.

Output Format: Winner name, Draw, or Invalid input.

Sample I/O: See repository README.

Primary sources

● https://github.com/tisha555/TCS_CodeVita_Season12

44. VIP Cafe - TCS CodeVita 2024 / Season 12

Synopsis. Simulate a dynamic priority queue of cafe orders and determine how many orders complete before a tracked customer.

Track: TCS CodeVita

Year / Season: 2024 / Season 12

Verification: Season 12 GitHub repository README lists it as a MockVita problem.

Difficulty Filter: Queue simulation with priority updates.

Constraints: Order count, priorities, and tracked position.

Input Format: N, list of N priorities, then tracked index K.

Output Format: Number of orders served before the tracked order.

Sample I/O: See repository README.

Primary sources

● https://github.com/tisha555/TCS_CodeVita_Season12

45. Weapon Boxes - TCS CodeVita 2024 / Season 12

Synopsis. Iteratively compare weapon-box weights, rotate lighter boxes, stop after K stable cycles, and compute labor cost excluding triangular-number weights.

Track: TCS CodeVita

Year / Season: 2024 / Season 12

Verification: Season 12 GitHub repository README lists it as a MockVita problem.

Difficulty Filter: Queue simulation plus numeric filtering.

Constraints: Box weights plus cycle-selection parameters N and K.

Input Format: Weights array, then N and K.

Output Format: Total labor cost after the process finishes.

Sample I/O: See repository README.

Primary sources

- https://github.com/tisha555/TCS_CodeVita_Season12

46. Zero Count - TCS CodeVita 2024 / Season 12

Synopsis. Place K ones inside a binary string of length L so that the longest zero-run is as short as possible.

Track: TCS CodeVita

Year / Season: 2024 / Season 12

Verification: Season 12 GitHub repository README lists it as a MockVita problem.

Difficulty Filter: Constructive minimization problem.

Constraints: $0 \leq K \leq L$ and L can be large on the linked sample page.

Input Format: Two integers, L and K.

Output Format: Length of the longest consecutive-zero block after optimal placement.

Sample I/O: Linked sample page includes examples for 3 1 -> 1 and 3 3 -> 0.

Primary sources

- https://github.com/tisha555/TCS_CodeVita_Season12
- <https://prepinsta.com/tcs-codevita/python-code-for-zero-count-tcs-codevita-prepinsta/>

47. On A Cube - TCS CodeVita 2025-2026 / Season 13

Synopsis. A beetle travels over a cube surface between honey spots, mixing same-face arc motion with cross-face shortest paths while avoiding the bottom face.

Track: TCS CodeVita

Year / Season: 2025-2026 / Season 13

Verification: Official TCS CodeVita sample page lists it as a current sample question.

Difficulty Filter: Geometric shortest-path / simulation problem.

Constraints: No point lies on the bottom face or on cube edges; $2 \leq N \leq 10$ on the official page.

Input Format: N followed by the ordered cube-surface coordinates of all visited points.

Output Format: Total traveled distance rounded to two decimal places.

Sample I/O: Official sample page includes example totals 4.05 and 6.05.

Primary sources

● <https://codevita.tcsapps.com/>

48. Sorting Boxes - TCS CodeVita 2025-2026 / Season 13

Synopsis. Sort parcel weights in increasing order except for a special placement of the heaviest box, minimizing pairwise swap effort defined by weight products.

Track: TCS CodeVita

Year / Season: 2025-2026 / Season 13

Verification: Official TCS CodeVita sample page lists it as a current sample question.

Difficulty Filter: Permutation / minimum-cost reordering problem.

Constraints: Distinct weights; cost of swapping two boxes is the product of their weights.

Input Format: N and office position k, followed by the box weights.

Output Format: Minimum total effort required.

Sample I/O: See official sample page.

Primary sources

● <https://codevita.tcsapps.com/>

49. Sport Stadium - TCS CodeVita 2025-2026 / Season 13

Synopsis. Seat a city contingent across alternating wet/dry seat blocks while minimizing span and respecting an upper bound on interior empty-seat gaps.

Track: TCS CodeVita

Year / Season: 2025-2026 / Season 13

Verification: Official TCS CodeVita sample page lists it as a current sample question.

Difficulty Filter: Greedy interval / feasibility optimization problem.

Constraints: Alternating dry/wet blocks, limited number of people willing to sit on wet seats.

Input Format: Official page describes seat-block counts and wet/dry block lengths.

Output Format: Minimum possible span between the first and last seated supporter.

Sample I/O: See official sample page.

Primary sources

● <https://codevita.tcsapps.com/>

50. Water Cistern - TCS CodeVita 2025-2026 / Season 13

Synopsis. Compute the shortest path for a bug moving on the accessible top and side surfaces of a cylinder from a starting point to a polar-style destination.

Track: TCS CodeVita

Year / Season: 2025-2026 / Season 13

Verification: Official TCS CodeVita sample page lists it as a current sample question.

Difficulty Filter: Surface-unfolding geometry problem.

Constraints: $40 < s \leq h < 10000$, $r < 100$, angle $0..359$ on the official page.

Input Format: Radius, height, start offset, then destination coordinates in the problem's custom coordinate system.

Output Format: Rounded shortest distance as an integer.

Sample I/O: Official sample page includes a cistern example.

Primary sources

- <https://codevita.tcsapps.com/>

51. Square Free Numbers - TCS CodeVita 2025-2026 / Season 13

Synopsis. Count the square-free divisors of N , excluding 1.

Track: TCS CodeVita

Year / Season: 2025-2026 / Season 13

Verification: Official TCS CodeVita sample page lists it as a current sample question.

Difficulty Filter: Number theory / divisor filtering problem.

Constraints: $N < 10^9$ and no prime factor larger than 19 on the official page.

Input Format: Single integer N .

Output Format: Count of square-free divisors greater than 1.

Sample I/O: Official sample page includes 20 -> 3 and 72 -> 3.

Primary sources

● <https://codevita.tcsapps.com/>

52. Codu and Sum Love - TCS CodeVita 2025-2026 / Season 13

Synopsis. Emulate the effect of a Java snippet over N input values, keep only the last two digits of each transformed number, then sum modulo 100.

Track: TCS CodeVita

Year / Season: 2025-2026 / Season 13

Verification: Official TCS CodeVita sample page lists it as a current sample question.

Difficulty Filter: Bitwise / modular-pattern reasoning problem.

Constraints: $1 \leq N \leq 10^7$ and $0 \leq x \leq 10^{18}$ on the official page.

Input Format: N, then N numeric values.

Output Format: Final code output after applying the snippet logic.

Sample I/O: Official sample page includes 8 6 7 4 -> 64.

Primary sources

● <https://codevita.tcsapps.com/>

53. Obstacle Game - TCS CodeVita 2025-2026 / Season 13

Synopsis. Follow the unique route from A to D in a grid and print the surrounding hurdle types at each route step.

Track: TCS CodeVita

Year / Season: 2025-2026 / Season 13

Verification: Official TCS CodeVita sample page lists it as a current sample question.

Difficulty Filter: Grid traversal / neighborhood inspection problem.

Constraints: $2 \leq N \leq 20$ with designated cell types A, D, R, S, L, W, T, M.

Input Format: N followed by the N x N character grid.

Output Format: Neighbor hurdles for each route step, then DESTINATION.

Sample I/O: Official sample page includes two complete grid examples.

Primary sources

● <https://codevita.tcsapps.com/>

54. Strange Algorithm String Conversion - TCS NQT Advanced Coding 2023

Synopsis. Convert string A into string B by repeatedly selecting a subset and collapsing it to the subset's lexicographically smallest character.

Track: TCS NQT Advanced Coding

Year / Season: 2023

Verification: PrepInsta Day 1 slot analysis includes the full story and examples for this advanced coding question.

Difficulty Filter: Advanced greedy / string transformation problem.

Constraints: $1 \leq N \leq 1000$; strings are lowercase and both have length N.

Input Format: N, then source string A, then target string B.

Output Format: Minimum move count, or -1 if conversion is impossible.

Sample I/O: Day 1 analysis includes examples such as de -> cd returning -1 and abab -> abaa returning 1.

Primary sources

- <https://prepinsta.com/tcs-nqt-all-slot-analysis-2023-day-1/>

55. Jersey Board Beats - TCS NQT Advanced Coding 2023

Synopsis. Students repeatedly move to positions indicated by fixed board numbers; compute how many beats are required to return everyone to the starting arrangement.

Track: TCS NQT Advanced Coding

Year / Season: 2023

Verification: PrepInsta Day 1 slot analysis includes the full story and examples for this advanced coding question.

Difficulty Filter: Permutation cycles / LCM-style reasoning problem.

Constraints: $1 \leq N \leq 100000$ and board values are a permutation of $1..N$.

Input Format: N followed by the board sequence.

Output Format: Number of beats needed to restore the original ordering.

Sample I/O: Day 1 analysis includes $1\ 2\ 3 \rightarrow 1$ and $2\ 3\ 1\ 5\ 4 \rightarrow 6$.

Primary sources

- <https://prepinsta.com/tcs-nqt-all-slot-analysis-2023-day-1/>

56. Golden House Exact-K Coins - TCS NQT Advanced Coding 2023

Synopsis. Choose an entry room and an exit room in a line of rooms so that the collected coin total is exactly K; at least one solution exists.

Track: TCS NQT Advanced Coding

Year / Season: 2023

Verification: PrepInsta Day 2 slot analysis includes the full story for this advanced coding question.

Difficulty Filter: Subarray / exact-sum selection problem.

Constraints: Source-linked only.

Input Format: Room counts, room coin values, and target K as described in the linked analysis.

Output Format: Required room selection or equivalent solution output from the original prompt.

Sample I/O: See linked 2023 Day 2 analysis.

Primary sources

● <https://prepinsta.com/tcs-nqt-all-slot-analysis-2023-day-2/>

57. Fair Sequence Maximum Sum - TCS NQT Advanced Coding 2023

Synopsis. Select a longest alternating-sign subsequence and maximize its total sum among all subsequences with that maximum length.

Track: TCS NQT Advanced Coding

Year / Season: 2023

Verification: PrepInsta Day 2 slot analysis includes the story and examples for this advanced coding question.

Difficulty Filter: Sequence DP / greedy sign-grouping problem.

Constraints: Array contains positive and negative values; sequence must alternate sign.

Input Format: Array length and array values.

Output Format: Maximum achievable sum for a fair subsequence of maximum length.

Sample I/O: See linked 2023 Day 2 analysis.

Primary sources

- <https://prepinsta.com/tcs-nqt-all-slot-analysis-2023-day-2/>

58. Unique Paths with Obstacles - TCS NQT Advanced Coding 2024 / 26 Apr Shift 1

Synopsis. Count paths from the top-left to bottom-right of a grid when moves are restricted to east or south and certain cells are blocked.

Track: TCS NQT Advanced Coding

Year / Season: 2024 / 26 Apr Shift 1

Verification: Campus Monk 2024 shift-wise post and a public GitHub shift repository both index this problem for 26 Apr 2024 Shift 1.

Difficulty Filter: Grid DP with obstacle handling.

Constraints: Campus Monk lists grid sizes up to 100 and K blocked cells.

Input Format: M N, then K, then K blocked-cell coordinates.

Output Format: Number of valid paths.

Sample I/O: See linked Campus Monk entry.

Primary sources

- <https://campusmonk.in/actually-tcs-coding-question-asked-shift-wise-2024-video-solution/>
- <https://github.com/Aditya-Mishra19/TCS-NQT-2024-Coding-Solutions/tree/main/26%20APRIL%202024%20SHIFT%20-%201/Unique%20Paths>

59. Subarray Equals Target - TCS NQT Advanced Coding 2024 / 26 Apr Shift 1

Synopsis. Find or count target-matching subarrays according to the 26 Apr Shift 1 prompt family.

Track: TCS NQT Advanced Coding

Year / Season: 2024 / 26 Apr Shift 1

Verification: Public GitHub shift repository indexes this problem under 26 Apr 2024 Shift 1.

Difficulty Filter: Target-sum subarray problem, included because it appears in the shift-specific advanced archive.

Constraints: Source-linked only.

Input Format: Source-linked only.

Output Format: Source-linked only.

Sample I/O: Source-linked only.

Archive Note: The shift/title is verified, but this compiled archive does not restate the full prompt text.

Primary sources

- <https://github.com/Aditya-Mishra19/TCS-NQT-2024-Coding-Solutions/tree/main/26%20APRIL%202024%20SHIFT%20-%201/Find%20subarray%20equals%20to%20target>

60. Sliding Window Maximum - TCS NQT Advanced Coding 2024 / 29 Apr Shift 1

Synopsis. Return the maximum element from every contiguous subarray of size K.

Track: TCS NQT Advanced Coding

Year / Season: 2024 / 29 Apr Shift 1

Verification: Campus Monk 2024 post and the public GitHub shift repository both index the maximum-in-every-subarray problem for this period.

Difficulty Filter: Deque / sliding-window maximum problem.

Constraints: Array length up to 10^5 in the Campus Monk write-up.

Input Format: Array values then window size K.

Output Format: Space-separated maxima for each size-K window.

Sample I/O: Campus Monk example uses array 1 4 7 7 6 8 3 with output 7 7 7 8.

Primary sources

- <https://campusmonk.in/actually-tcs-coding-question-asked-shift-wise-2024-video-solution/>
- <https://github.com/Aditya-Mishra19/TCS-NQT-2024-Coding-Solutions/tree/main/29%20APRIL%202024%20Shift%20-%201/Maximum%20element%20of%20subarray>

61. Distinct Bitwise-OR Subarray Results - TCS NQT Advanced Coding 2024 / 29 Apr Shift 2

Synopsis. Compute how many distinct values can be produced by taking the bitwise OR of every possible subarray.

Track: TCS NQT Advanced Coding

Year / Season: 2024 / 29 Apr Shift 2

Verification: Campus Monk 2024 post and the public GitHub shift repository both index the Bit OR Sum problem.

Difficulty Filter: Set-based rolling bitwise DP problem.

Constraints: Campus Monk lists N up to 100 and values below 10^9 .

Input Format: One line containing the array values.

Output Format: Count of distinct OR values across all subarrays.

Sample I/O: Campus Monk example uses 1 2 3 and returns 3.

Primary sources

- <https://campusmonk.in/actually-tcs-coding-question-asked-shift-wise-2024-video-solution/>
- <https://github.com/Aditya-Mishra19/TCS-NQT-2024-Coding-Solutions/tree/main/29%20APRIL%20SHIFT%202/Bit%20OR%20Sum>

62. Ordered Maximum Difference - TCS NQT Advanced Coding 2024 / 03 May Shift 1

Synopsis. Return the largest difference between a later larger number and an earlier smaller number while respecting array order.

Track: TCS NQT Advanced Coding

Year / Season: 2024 / 03 May Shift 1

Verification: Campus Monk 2024 post and the public GitHub shift repository both index this problem for the early-May shifts.

Difficulty Filter: Linear scan with running minimum.

Constraints: Source-linked only.

Input Format: n followed by n integers.

Output Format: Maximum valid difference or 0 if no increasing pair exists.

Sample I/O: Campus Monk example uses -3 -5 1 6 -7 8 11 and returns 18.

Primary sources

- <https://campusmonk.in/actually-tcs-coding-question-asked-shift-wise-2024-video-solution/>
- <https://github.com/Aditya-Mishra19/TCS-NQT-2024-Coding-Solutions/tree/main/3%20May%20Shift%201/maximum%20difference%20between%20smallest%20and%20largest>

63. Maximum Product of Three Efficiencies - TCS NQT Advanced Coding 2024 / 03 May Shift 2

Synopsis. Choose any three employee efficiencies, including negative values when useful, to maximize the product.

Track: TCS NQT Advanced Coding

Year / Season: 2024 / 03 May Shift 2

Verification: Campus Monk 2024 post and the public GitHub shift repository both index the Maximum Efficiency problem.

Difficulty Filter: Order-statistics / signed-product optimization problem.

Constraints: $3 \leq n \leq 1000$ and values can be negative.

Input Format: n followed by n efficiency values.

Output Format: Maximum product achievable using exactly three values.

Sample I/O: See linked Campus Monk write-up.

Primary sources

- <https://campusmonk.in/actually-tcs-coding-question-asked-shift-wise-2024-video-solution/>
- <https://github.com/Aditya-Mishra19/TCS-NQT-2024-Coding-Solutions/tree/main/3%20May%20SHIFT%202/Maximum%20Efficiency>

64. Split Array with Equal Averages - TCS NQT Advanced Coding 2025 / 03 Oct Shift 2

Synopsis. Check whether an integer array can be split into two non-empty contiguous parts that share the same average.

Track: TCS NQT Advanced Coding

Year / Season: 2025 / 03 Oct Shift 2

Verification: Campus Monk's 2025 shift-wise post indexes this as an actual TCS NQT coding question.

Difficulty Filter: Prefix-sum / ratio-equality reasoning problem.

Constraints: Source-linked only.

Input Format: Array length and array values.

Output Format: true or false depending on whether such a split exists.

Sample I/O: See linked Campus Monk 2025 article.

Primary sources

- <https://campusmonk.in/tcs-nqt-2025-actual-coding-questions-with-shift-wise-java-solution/>

65. Jump Game Reachability - TCS NQT Advanced Coding 2025 / 04 Oct Shift 2

Synopsis. Determine whether the last index can be reached when each array value gives the maximum jump length from that position.

Track: TCS NQT Advanced Coding

Year / Season: 2025 / 04 Oct Shift 2

Verification: Campus Monk's 2025 shift-wise post indexes this as an actual TCS NQT coding question.

Difficulty Filter: Greedy reachability problem.

Constraints: Source-linked only.

Input Format: Comma-separated jump lengths on the linked write-up.

Output Format: Boolean reachability result.

Sample I/O: See linked Campus Monk 2025 article.

Primary sources

- <https://campusmonk.in/tcs-nqt-2025-actual-coding-questions-with-shift-wise-java-solution/>